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From CAGR to CPCV: A Framework for Robust XAUUSD Strategy Calibration Using Advanced Metrics and Multi-Objective Optimization

This report provides a deep research analysis into alternative scoring mechanisms and optimization frameworks for the calibration of the XAUUSD v6 algorithmic trading system. The central challenge addressed is the unreliability of Compound Annual Growth Rate (CAGR) as a performance metric for micro-trials, which lack sufficient historical data for stable estimation . The investigation explores a spectrum of industry-standard solutions, ranging from enhanced single-objective functions with better-behaved gradients to sophisticated multi-objective frameworks and robust validation methodologies. The analysis draws upon quantitative finance literature on market microstructure and portfolio optimization, as well as machine learning research on surrogate modeling and sample-efficient optimization. The objective is to identify metrics that maximize information density for samplers, maintain distinct strategy quality signals, and ultimately inform a redesign of the executable objective function currently based on CAGR minus linear penalties.

Fundamental Limitations of CAGR-Based Objectives in Low-Sample Regimes

The reliance on Compound Annual Growth Rate (CAGR) as a primary optimization objective, especially when augmented with linear penalties for constraint violations, presents significant conceptual and practical flaws within the context of micro-trial calibration for the XAUUSD v6 system. This approach, while seemingly straightforward, inherits the fundamental weaknesses of its conceptual predecessors, most notably the standard Sharpe Ratio, which it implicitly aims to annualize and optimize [13]. These limitations are not merely theoretical; they directly undermine the reliability of the Bayesian optimization process, leading to the selection of parameters that may appear

optimal on a limited backtest but are likely overfitted to noise rather than genuine market structure. The core issues stem from three interconnected areas: the violation of foundational statistical assumptions, insensitivity to asymmetric risk profiles, and a high susceptibility to selection bias in low-sample environments.

First, the entire framework of mean-variance analysis, upon which the Sharpe Ratio and its derivatives like CAGR are built, rests on the assumption that financial returns are independently and identically distributed (IID) and follow a normal distribution [13] [25]. This assumption is demonstrably false for real-world financial time series, including those generated by algorithmic trading strategies. Financial data exhibits strong serial correlation, where sequential observations are dependent on preceding values, and volatility clustering, where periods of high volatility are followed by more high volatility and vice versa [25] [32]. Applying standard k-fold cross-validation, which assumes IID data, to financial time series introduces severe information leakage, where future data inadvertently influences the training process, artificially inflating performance estimates [2] [28]. This leads to overly optimistic results that fail to generalize to out-of-sample data [25]. While walk-forward analysis attempts to correct for this temporal dependency, a simple point estimate like CAGR derived from a single, albeit rolling, window remains vulnerable to the same underlying statistical issues [46]. The CAGR calculation itself is susceptible to distortions from non-normal return distributions, particularly those with significant skewness (asymmetry) and excess kurtosis (fat tails), which are common in algorithmic strategy returns [32].

Second, the standard Sharpe Ratio treats all volatility equally, penalizing both upside and downside deviations from the mean [32]. This symmetric treatment of risk is economically unsound and often misleading for evaluating trading strategies, which frequently exhibit asymmetric return profiles. For example, a trend-following strategy might generate large positive returns with correspondingly large positive volatility, while simultaneously having very small negative returns during losing periods. The standard Sharpe Ratio would penalize the large upside volatility, potentially undervaluing such a strategy. Metrics like the Sortino Ratio address this by replacing the standard deviation in the denominator with *downside deviation*, which only considers the volatility of returns falling below a Minimum Acceptable Return (MAR), typically the risk-free rate or zero [32]. Similarly, the Omega Ratio offers a more holistic view by considering the entire shape of the return distribution relative to a threshold, rather than just the first two moments (mean and variance) [30] [32]. By focusing solely on CAGR, the current objective function fails to distinguish between profitable growth achieved with controlled downside risk and growth driven by extreme, volatile swings that expose the capital

curve to unacceptable drawdowns. It lacks any inherent mechanism to penalize tail risk or asymmetrical return characteristics.

Third, and perhaps most critically for the problem at hand, is the profound vulnerability to selection bias and overfitting in low-sample, high-dimensional search spaces. The Bayesian optimization process in Optuna systematically explores a vast parameter space, effectively performing a massive number of trials to find an optimum [29]. In a micro-trial environment with limited data points, it becomes statistically inevitable that some random combination of parameters will produce a spuriously high CAGR purely by chance [19]. This phenomenon is known as selection bias or data snooping. Even if all tested strategies have zero expected return (i.e., no skill), the maximum observed Sharpe ratio (and by extension, CAGR) from a large cohort of trials will be positive due to sampling noise alone [19]. For instance, testing as few as 50 independent strategies could lead to an expected spurious maximum Sharpe ratio of 0.4-0.8, easily mistaken for a real edge [19]. The current objective, `cagr - linear_penalties`, does little to combat this. While linear penalties for max drawdown or trade frequency can act as a weak regularizer, they are applied after the fact and may not be sufficient to reject a configuration that has found a lucky alignment of parameters producing a high CAGR on a short track record. The problem is exacerbated by the fact that many strategies may have very irregular trading frequencies, making a simple count of trades per day an imperfect proxy for robustness [11]. The linear penalty structure implies a uniform cost for violating a constraint, whereas the economic impact of, for example, exceeding a max drawdown limit by 1% versus 5% is not linear. This highlights the need for a more principled approach to balancing multiple objectives and penalizing undesirable outcomes. The use of a simple CAGR score provides minimal information to the sampler about the nature of the performance it has achieved, making it difficult for the optimizer to learn the true underlying relationship between parameters and robust, generalizable performance.

Enhanced Single-Objective Optimization: Differentiable Surrogates and Composite Functions

To overcome the limitations of the CAGR-based objective, a more effective strategy involves replacing it with a single, more sophisticated objective function that provides a richer, more informative signal to the Bayesian optimizer. This pathway focuses on two main classes of alternatives: differentiable surrogates designed to guide the sampler more

efficiently, and composite objective functions engineered to inherently penalize overfitting and undesirable risk characteristics. Both approaches aim to transform the optimization landscape into one that is more conducive to finding robust, generalizable solutions.

Differentiable surrogates represent a powerful enhancement because they allow the optimizer to leverage gradient-like information, even if the underlying function is not analytically differentiable. Modern Bayesian optimizers, including those implemented in libraries like Optuna, use probabilistic models (e.g., Gaussian Processes) to approximate the objective function and acquisition functions (like Expected Improvement) that balance exploration and exploitation [37] [38]. A smooth, differentiable objective function provides a clearer signal for this approximation, leading to more efficient convergence. Several key metrics from quantitative finance have been formulated as smooth, differentiable functions suitable for this purpose. The **Sortino Ratio**, which uses downside deviation in the denominator, is a prime candidate [32] [36]. Its focus on unfavorable volatility aligns perfectly with the goal of capital preservation. Research has demonstrated the feasibility of creating smooth, differentiable versions of the Sortino ratio, which can be used directly as an objective or evaluation metric in optimization loops [30]. Similarly, the **Omega Ratio**, which considers the entire return distribution, has also been expressed in a differentiable form, offering an even more comprehensive measure of risk-adjusted performance [30]. Optimizing for a differentiable surrogate like the Sortino or Omega ratio would provide the Optuna sampler with a much clearer direction than CAGR, explicitly rewarding strategies that generate returns with controlled downside risk.

Beyond individual metrics, composite objective functions offer a way to bake multiple desirable properties directly into a single score. The most prominent example discussed in the provided literature is the **GT-Score**, a metric designed specifically to reduce overfitting in trading strategy optimization [3]. The GT-Score formula, $GT\text{-}Score = (\mu_s - \mu_b) \cdot \ln(z) \cdot R^2 / \sigma_d$, integrates several critical components: it includes a term for statistical significance ($\ln(z)$), a measure of consistency (R^2), and a denominator based on downside deviation (σ_d) [3]. In empirical studies, the GT-Score demonstrated superior generalization ability, achieving a generalization ratio nearly twice that of baseline methods like the Sharpe Ratio [3]. While it adds computational overhead, its explicit design for combating overfitting makes it highly relevant for the XAUUSD micro-trial problem. Another compelling source of inspiration comes from research on training neural networks for trading, which proposes a suite of finance-grounded, differentiable loss functions [48]. These include `SharpeLoss`, `LogMDDLoss` (a smooth variant of Maximum Drawdown), and various composite losses that combine

them, such as `RiskAdjLoss` [48]. Empirical tests showed that an LSTM model trained with `LogMDDLoss` achieved a Sharpe ratio of 1.7573, whereas the same model trained with a standard Mean Squared Error (MSELoss) objective produced a negative Sharpe ratio of -0.4564, demonstrating the profound impact of aligning the training objective with the final performance metric [48].

A practical implementation of this approach would involve modifying the `objective_wrapper` function in the calibrators (`optuna_calibrate_*.py`). Instead of calculating a single CAGR value, the wrapper would compute a more robust metric. For example, it could calculate the median Sharpe Ratio across multiple CV folds or, more powerfully, the Probabilistic Sharpe Ratio (PSR), which accounts for non-normality [13] [42]. The wrapper would then return this new, more informative scalar value as the objective score for Optuna. This change requires careful consideration of normalization. If combining multiple metrics into a single composite score, each component should be normalized to a comparable scale to prevent one dominant metric from overwhelming the others [43]. Techniques like logit transformations can be used to map plausible ranges of each metric to a [0, 1] interval before summing them with calibrated weights [43]. This allows the optimizer to explore the trade-offs between, for instance, profit factor and trade count in a more balanced manner. The table below compares potential single-objective replacements for the current CAGR-based score.

Metric/Objective	Core Principle	Advantages for XAUUSD Calibration	Key Considerations
Sortino Ratio	Replaces standard deviation with downside deviation in the denominator [32].	Explicitly penalizes downside risk, providing a clearer signal for drawdown-sensitive strategies. Smooth, differentiable versions exist [30].	Assumes a Minimum Acceptable Return (MAR); may require tuning the MAR threshold.
Probabilistic Sharpe Ratio (PSR)	Calculates the probability that an observed Sharpe ratio is statistically significant, accounting for non-normality and sample size [12] [13].	Directly addresses the non-normal, fat-tailed nature of trading returns. Attemporal by nature, suitable for irregular trades [13].	Requires sufficient data for reliable Z-score calculation; sensitive to skewness and kurtosis estimates [3].
GT-Score	Composite score combining statistical significance ($\ln(z)$), consistency (R^2), and downside risk (σ_d) [3].	Explicitly designed to reduce overfitting and improve generalization. Proven to retain more training performance out-of-sample [3].	Higher computational complexity than simpler ratios. Requires a benchmark return (μ_b).
Differentiable MDD Loss (`LogMDDLoss`)	Smooth, differentiable formulation of Maximum Drawdown as a loss function [48].	Provides a direct, powerful penalty for large drawdowns, which is crucial for capital preservation. Aligns training with final PnL.	Can be highly non-convex, potentially creating a challenging optimization landscape.
Composite Risk-Adjusted Score	Sum of multiple normalized metrics (e.g., PSR, Profit Factor, `-MaxDD`) with calibrated weights [43].	Holistic view of performance, allowing explicit control over the trade-off between return, risk, and consistency.	Requires careful tuning of weights. Normalization is critical to avoid metric dominance [43].

By adopting one of these enhanced single-objective functions, the calibration process can move beyond the simplistic and flawed premise of maximizing CAGR. It would instead seek parameters that yield statistically significant, downside-aware, and robust performance, providing the Optuna sampler with a far more informative and reliable gradient to follow.

Multi-Objective Optimization: Preserving Signal Dimensions via Pareto Front Analysis

An alternative and more philosophically rigorous approach to replacing the single CAGR objective is to abandon the single-score paradigm entirely and adopt a Multi-Objective Optimization (MOO) framework. Instead of forcing all competing goals—such as maximizing returns, minimizing drawdown, and ensuring adequate trade frequency—into a single composite score, MOO seeks to optimize them simultaneously. This method preserves the separate dimensions of strategy performance, revealing the inherent trade-offs between them and presenting a decision-maker with a set of optimal compromises known as the Pareto front [14]. This approach is particularly well-suited for complex problems like trading strategy development, where no single solution can be optimal across all desired metrics [14] [43].

The core principle of MOO is to find a set of non-dominated solutions. A solution is Pareto-optimal if it is impossible to improve one objective without worsening at least one other [7] [14]. For example, in a two-objective problem of maximizing Sharpe Ratio and minimizing Maximum Drawdown, a point on the Pareto front represents a strategy where any attempt to lower the maximum drawdown would necessarily result in a lower Sharpe ratio, and vice-versa. The collection of all such non-dominated solutions forms the Pareto front, which visually represents the best possible achievable trade-offs between the conflicting objectives [10] [14]. This stands in stark contrast to the single-objective approach, which implicitly forces the decision-maker's preferences into the weighting scheme of the composite function, potentially hiding important trade-offs or misrepresenting the true landscape of possibilities.

The leading algorithm for solving MOO problems is the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [16]. NSGA-II is a popular multi-objective evolutionary algorithm that incorporates elitism, fast non-dominated sorting to rank solutions, and a crowding distance mechanism to maintain diversity along the Pareto front [16] [18]. Its design avoids

the need for user-defined weights to trade off between objectives, making it intuitive to apply [18]. The algorithm maintains a population of candidate solutions and iteratively evolves it through selection, crossover, and mutation. In each generation, the population is sorted into non-dominated fronts, with the first front containing the best (non-dominated) solutions, the second front containing the next-best, and so on. Selection proceeds by filling the next generation starting from the first front, moving to the second, etc., until the population size is reached [16]. To manage the selection within a front, NSGA-II uses a crowding distance metric, which promotes solutions that are spread out and isolated from their neighbors, thus preserving diversity across the Pareto front [16]. Libraries like Pymoo provide robust implementations of NSGA-II, making it accessible for integration into Python-based workflows [8].

Implementing an MOO framework for the XAUUSD system would require a significant architectural shift. The `objective_wrapper` function in the calibrators would need to be modified to return a tuple or list of scores, such as `[annualized_return, -max_drawdown, sharpe_ratio, profit_factor]`. The existing Optuna Bayesian optimization infrastructure would need to be replaced with an MOO algorithm like NSGA-II. This transition moves the optimization from a gradient-based search guided by a surrogate model to an evolutionary search guided by Pareto dominance. Given the high computational cost of evaluating a trading strategy, a direct application of NSGA-II could be prohibitive, as it typically requires a large number of function evaluations to converge [7]. To address this, the framework can be augmented with surrogate-assisted evolutionary algorithms (SAEAs) [7]. In this hybrid approach, expensive-to-evaluate objective functions are approximated by cheaper surrogate models, such as Gaussian Processes (GPs) [5] [8]. The evolutionary algorithm performs most of its operations using the fast surrogate models, only calling the full backtester for a select number of promising candidate solutions identified by an acquisition function [8]. This dramatically reduces the number of costly backtests required, making MOO computationally feasible for problems like strategy calibration. Adaptive schemes can dynamically update the surrogate models with new data points, improving accuracy in promising regions of the search space [8].

However, MOO is not without its own set of challenges and limitations. One major limitation of NSGA-II is its performance degradation when the number of objectives increases to three or more [17]. The crowding distance mechanism, which works well in two or three dimensions, becomes less effective, leading to poor coverage of the Pareto front and inefficient search [17] [18]. For problems with many objectives, alternative algorithms like NSGA-III or SMS-EMOA, which use different diversity maintenance

mechanisms, may be more appropriate [17]. Furthermore, MOO methods are generally less sample-efficient than single-objective Bayesian optimization, meaning they may require more iterations to find a satisfactory solution [7]. Finally, interpreting the results requires a different mindset. The goal shifts from selecting a single "best" trial to analyzing the entire Pareto front and choosing a solution based on qualitative or operational criteria, such as a target level of risk aversion or a desired turnover rate [14]. This places a greater analytical burden on the developer but provides a much deeper understanding of the strategy's performance landscape. The table below outlines the key differences between the single-objective and MOO paradigms.

Aspect	Single-Objective Optimization (Current Approach)	Multi-Objective Optimization (Proposed Approach)
Objective Function	Returns a single scalar value (e.g., `CAGR - penalties`).	Returns a vector of scores (e.g., `[return, -drawdown, sharpe]`).
Optimization Goal	Find a single set of parameters that maximizes/minimizes one score.	Find a set of Pareto-optimal solutions representing the best trade-offs.
Handling Trade-offs	Implicitly handled by the weights in a composite score, which can be arbitrary.	Explicitly revealed by the shape of the Pareto front.
Primary Algorithm	Bayesian Optimizers (e.g., Optuna with GP-EI).	Evolutionary Algorithms (e.g., NSGA-II).
Information Provided	A single "best" parameter set and its corresponding score.	A diverse archive of high-quality, non-dominated solutions.
Decision-Making Process	Select the trial with the highest (or lowest) objective value.	Analyze the Pareto front to select a solution that meets specific risk/reward criteria.
Computational Cost	Generally lower; BO is sample-efficient.	Generally higher; requires more function evaluations, though SAEAs can mitigate this [7].
Key Challenge	Choosing appropriate weights for a composite score; risk of misrepresenting the true trade-offs.	Managing the "curse of dimensionality" (performance degrades with >3 objectives) [17], and interpreting the Pareto front.

In summary, transitioning to a MOO framework represents a significant step up in sophistication for strategy calibration. It abandons the flawed notion of a single "best" strategy in favor of a comprehensive exploration of the performance landscape. While requiring substantial engineering effort, it offers the deepest insights into the trade-offs inherent in the XAUUSD strategies and provides a robust portfolio of high-performing parameter sets.

Robust Validation Frameworks: Mitigating Overfitting with Time-Series Cross-Validation

While enhancing the objective function is crucial, an equally important aspect of improving micro-trial calibration is fundamentally changing how performance is measured *within* each trial. The core problem with relying on a single CAGR figure is its inherent unreliability in a low-sample regime; it is a point estimate prone to being a statistical fluke. The most effective defense against this is to replace a single backtest with a robust validation framework that generates a distribution of out-of-sample performance metrics. Among the techniques available, Combinatorial Purged Cross-Validation (CPCV) stands out as a state-of-the-art methodology specifically designed for financial time-series data, championed by Marcos López de Prado [1] [27].

Standard cross-validation techniques are ill-suited for time-series data because they violate the assumption of independence by allowing information from the test set to leak into the training set [25]. CPCV directly addresses this by generating a multitude of chronologically consistent train-test partitions and systematically purging overlapping information between them [1]. The process works by partitioning the dataset into contiguous blocks and then combinatorially generating numerous distinct out-of-sample paths, where each path consists of a sequence of non-overlapping training and testing windows [27]. Before each fold, a 'purge' operation removes any training observations whose labels overlap in time with the upcoming test set, eliminating look-ahead bias [24] [28]. An 'embargo' period is then applied immediately following the test set, during which adjacent data points are excluded from both subsequent training sets and the test set itself [25] [26]. This embargo period is crucial as it prevents information from propagating from the test period into the training process, even for features with serial correlation [28]. The result is not a single performance score, but an empirical distribution of out-of-sample outcomes, providing a much more realistic and robust assessment of a strategy's potential generalization ability [1].

The philosophical shift enabled by CPCV is profound. Instead of seeking a single 'optimal' parameter set that happens to perform best on one specific historical path, the goal becomes identifying a 'robust parameter region' that performs consistently well across hundreds or thousands of counterfactual scenarios [1]. This reframes the optimization task from finding a peak to finding a plateau—a region in the parameter space that yields good performance under varying conditions. This approach directly counters the problem of selection bias, as a parameter set that survives scrutiny across numerous independent OOS paths is far less likely to be a lucky coincidence [1]. The `purgedcv` Python library

provides the necessary tools to implement this methodology, including `CombinatorialPurgedCV` and helper functions to reconstruct the full backtest paths from the individual folds [27]. Critically, this library also includes integrations for optimization frameworks, such as the `TrialSharpeRecorder` for Optuna, which can automatically collect the Sharpe ratios from each fold, providing the necessary data for further statistical analysis [27].

Integrating CPCV into the XAUUSD calibration pipeline would involve modifying the `backtest_xauusd_signal_csv.py` script. The current logic that runs a single backtest for a given set of parameters would be replaced by an outer loop that iterates through the splits generated by `CombinatorialPurgedCV`. Inside this loop, the strategy would be trained on the designated in-sample data and evaluated on the out-of-sample data. The resulting performance metrics (e.g., Sharpe ratio, CAGR, max drawdown) from each fold would be collected. After the loop completes, a final aggregate score would be calculated from the distribution of fold-level metrics. Instead of taking a simple mean, which can be skewed by outliers, more robust statistics like the median are recommended [42]. For example, the final score returned to Optuna could be the median Sharpe ratio across all folds, or a more advanced metric like the Probabilistic Sharpe Ratio (PSR) calculated from the distribution of Sharpe ratios [13] [27]. This ensures that a parameter set with one spectacularly lucky fold and many mediocre ones is not unfairly favored over a set with consistently moderate performance across all folds.

Other related techniques, such as Purged K-Fold CV and Walk-Forward Analysis (WFA), share the same core principles of preventing information leakage and ensuring out-of-sample validity [24] [45]. WFA, in particular, is a widely accepted industry practice involving a rolling window of in-sample optimization followed by out-of-sample testing [45] [46]. The Walk-Forward Ratio (WFR), defined as the ratio of the out-of-sample Sharpe to the average in-sample Sharpe, is a key metric for quantifying a strategy's generalization ability [47]. A WFR value greater than 0.5 is often considered indicative of a well-generalized strategy [47]. Combining these methods provides a comprehensive toolkit for robust validation. However, CPCV is arguably the most rigorous of these, as it systematically explores a vast number of possible train-test configurations, providing a more thorough examination of a strategy's robustness than a single WFA run [45]. The table below summarizes the key validation techniques and their relevance.

Technique	Primary Goal	Mechanism	Suitability for XAUUSD Micro-Trials
Standard K-Fold CV	Estimate model performance.	Randomly splits data into k folds, cycling through each as a test set.	Unsuitable. Violates temporal dependencies and causes severe information leakage in time-series data [25] [28].
Purged K-Fold CV	Prevent information leakage in time-series.	Chronological split with a 'purge' to remove overlapping labels and an 'embargo' buffer post-test [24] [28].	Highly Suitable. A foundational technique for any time-series validation. Integrates well with libraries like `purgedcv` [27].
Walk-Forward Analysis (WFA)	Simulate live trading conditions.	Rolling window of in-sample optimization followed by out-of-sample testing [45] [47].	Standard Industry Practice. Excellent for simulating real-world deployment but may still be susceptible to selection bias if not paired with CPCV [45].
Combinatorial Purged CV (CPCV)	Identify robust parameter *regions* [1].	Generates C(N,K) folds to create multiple distinct OOS backtest paths, providing a distribution of performance metrics [27].	Most Suitable. Provides the most rigorous assessment of generalization by exposing parameters to countless counterfactual histories. The gold standard for avoiding overfitting [1].

In conclusion, implementing a robust validation framework like CPCV is arguably the single most impactful change that can be made to the calibration process. It directly confronts the root cause of the CAGR problem—the unreliability of a single performance estimate—by providing a distribution of outcomes. This aligns perfectly with the goal of maximizing information density for the sampler and building confidence that the selected parameters represent a genuine signal rather than a statistical artifact.

Advanced Statistical Metrics for Significance Testing and Selection Bias Correction

Beyond simply changing the validation methodology, a more nuanced approach involves integrating advanced statistical metrics directly into the objective function. These metrics go a step further than traditional ratios by quantifying the statistical significance of a strategy's performance and correcting for the pervasive issue of selection bias inherent in extensive backtesting. Two of the most prominent metrics in this category, proposed by Bailey and López de Prado, are the Probabilistic Sharpe Ratio (PSR) and the Deflated Sharpe Ratio (DSR) [13] [19].

The **Probabilistic Sharpe Ratio (PSR)** was developed to address the primary weakness of the standard Sharpe ratio: its reliance on the assumption of normally distributed returns [13]. Since financial returns are often non-normal, exhibiting skewness and excess kurtosis, the standard SR can be misleading. The PSR calculates the probability that an observed Sharpe ratio ($\hat{SR}$) is actually greater than a given benchmark Sharpe ratio

(SR^*), taking into account the sample size, skewness ($\hat{\gamma}_3$), and kurtosis ($\hat{\gamma}_4$) of the return stream [12] [13]. The formula is complex, adjusting the Z-statistic used in hypothesis testing to account for these higher moments, thereby providing a more accurate measure of confidence in the reported Sharpe ratio [13]. For the XAUUSD system, this is particularly valuable because strategy returns can be highly non-normal, and the trading frequency can be irregular, making annualization questionable [13]. The PSR is an atemporal measure, calculated on the original data frequency, which makes it robust to these variations [13]. When used within a CPCV framework, one can calculate the PSR for the Sharpe ratio observed in each fold, or even calculate a single PSR from the entire distribution of out-of-sample Sharpe ratios. The `purgedcv` library provides a direct implementation of the PSR calculation, facilitating its integration into the calibration workflow [27].

Building upon the PSR, the **Deflated Sharpe Ratio (DSR)** tackles the second major issue: selection bias [19]. When optimizing a strategy over a large parameter space, one is inevitably testing many "ghost" strategies that are never deployed. Even if all these strategies have zero true skill, the one that appears to have the highest Sharpe ratio will be positive due to chance alone [19]. The DSR corrects for this by asking a harder question: "What is the probability that the true Sharpe ratio exceeds zero, given that we selected the best of K strategies?" [19]. It uses extreme value theory to adjust the null hypothesis from $H_0: SR = 0$ to $H_0: SR = E[\max\{SR\}]$, the expected maximum Sharpe from K unskilled trials [19]. This makes it much harder for a spurious result to pass the test. The DSR is presented as a superior metric because it explicitly penalizes overfitting by counting every parameter variation as a trial [20]. For the XAUUSD calibration, where Optuna explores a vast hyperparameter space, the DSR provides a powerful tool for distinguishing genuine edges from artifacts of the search process. The interpretation is straightforward: a DSR value of 0.95 or higher suggests the strategy survives multiple testing at the 5% level, while a value below 0.5 indicates it is likely explained by selection bias and should be rejected [19].

Despite their conceptual strength, these metrics are not without controversy. One critique argues that the DSR is statistically invalid and suffers from extremely low power, meaning it is overly conservative and may fail to detect true positive signals [21]. The author contends that the DSR's derivation relies on indefensible assumptions about the normality of Sharpe ratios and that the statistic is not uniformly distributed under the null hypothesis, leading to incorrect Type I error rates [21]. This criticism highlights the ongoing debate in the field and suggests that while these metrics are powerful concepts, their practical application requires careful consideration of their assumptions and

limitations. Nonetheless, for the purpose of this research, their inclusion as potential objective functions is warranted.

A third category of advanced metrics involves **information-theoretic acquisition functions** for Bayesian optimization itself. Rather than optimizing a traditional performance metric, these methods aim to maximize the information gained about the location of the global optimum in each iteration [39]. Methods like Entropy Search (ES) and Max-value Entropy Search (MES) model the uncertainty about the optimizer and select the next evaluation point to minimize this uncertainty most rapidly [39]. While computationally intensive, they represent a frontier in BO that could lead to faster convergence. For instance, MES focuses on gaining information about the value of the objective function at the optimum, which can be more tractable than tracking the location of the optimum itself [39]. These methods could be particularly useful in later stages of optimization after a coarse search has narrowed down the promising regions of the parameter space.

The following table compares these advanced statistical metrics:

Metric/Method	Core Principle	Strengths for XAUUSD Calibration	Weaknesses & Considerations
Probabilistic Sharpe Ratio (PSR)	Probability that an observed SR is greater than a benchmark, adjusted for non-normality (skew/kurtosis) and sample size [12] [13].	Directly addresses non-normal returns common in trading. Atemporal nature is suitable for irregular trading frequencies. Library support exists [27].	Requires reliable estimates of skewness and kurtosis, which can be noisy in small samples. Still susceptible to selection bias if used naively.
Deflated Sharpe Ratio (DSR)	Corrects the PSR for selection bias from multiple testing (the "look-elsewhere effect") [19].	Explicitly penalizes overfitting by accounting for the number of trials in the optimization process. Strong conceptual fit for Bayesian optimization [20].	Controversial; criticized as statistically invalid and overly conservative [21]. Requires an accurate count of independent trials. Unreliable with very small datasets ($T < 50$) [19].
Information-Theoretic Acquisition	Selects next evaluation point to maximize information gain about the location/value of the global optimum [39].	More theoretically rigorous than local criteria like Expected Improvement (EI). Could achieve faster convergence in later optimization stages.	Computationally very expensive (e.g., Monte Carlo sampling, variational inference). May be too complex for initial implementation. Primarily a research-grade technique.

Incorporating these advanced metrics would elevate the calibration process from simple performance ranking to a rigorous statistical inference problem. By optimizing for the PSR or DSR, the objective function itself becomes a test of statistical significance and generalizability. This aligns perfectly with the project's goal of replacing a fragile CAGR-based score with something more robust. The most pragmatic starting point would be to calculate the PSR from the distribution of out-of-sample results generated by CPCV and use that as the final score for the optimizer.

Synthesis and Phased Implementation Roadmap

The investigation into industry-standard scoring metrics and optimization frameworks reveals a clear path forward for replacing the CAGR-based objective in the XAUUSD v6 system. The initial approach, reliant on a single, unstable metric in a low-sample regime, is fundamentally flawed. The solution lies in a multi-faceted strategy that combines robust validation, superior scoring metrics, and, eventually, more sophisticated optimization paradigms. The directive to "try everything" can be satisfied through a phased implementation roadmap that prioritizes high-impact changes with manageable effort, gradually progressing towards more complex but powerful solutions.

Phase 1: Immediate Foundation - Implementing Robust Validation and a Superior Score

The highest priority is to establish a foundation of statistical rigor. The single most impactful change is to replace the single backtest with a robust validation framework. 1. **Action:** Integrate **Combinatorial Purged Cross-Validation (CPCV)** into the backtesting loop within `backtest_xauusd_signal_csv.py`. Utilize the `purgedcv` Python library to generate multiple independent out-of-sample paths [1] [27]. This will transform the evaluation of each parameter set from a single-point estimate into a distribution of performance metrics. 2. **Action:** Within the CPCV framework, replace the final aggregation of the CAGR score with a more robust metric. The most straightforward yet effective choice is the **Probabilistic Sharpe Ratio (PSR)**, calculated from the distribution of out-of-sample Sharpe ratios generated by the folds [13] [27]. Alternatively, a simpler but still effective option is the **Sortino Ratio**, calculated from the final OOS equity curve [32]. 3. **Rationale:** This phase directly addresses the core problem of unreliable micro-trial data. CPCV mitigates information leakage and provides a distribution of outcomes, making the results far less susceptible to luck [1]. Optimizing for the PSR or Sortino ratio instead of CAGR provides a stronger, more economically meaningful signal to the sampler, explicitly rewarding statistically significant, downside-controlled performance [13] [32]. This combination represents a substantial upgrade in quality with a relatively contained scope of implementation.

Phase 2: Intermediate Refinement - Enhancing Objectives and Constraint Handling

Once a robust validation framework is in place, the next step is to refine the objective function itself. 1. **Action:** Develop a custom **composite objective function** that synthesizes multiple metrics into a single score. Inspired by the GT-Score, this function could combine the PSR, a normalized Profit Factor, and a penalty for low trade counts [3]

[43]. The weights for this composite score could themselves be tuned using a secondary layer of optimization. 2. **Action:** Leverage **Optuna's built-in constraint handling** capabilities. Keep the primary objective as CAGR or the PSR, but add hard constraints for risk metrics. For example, define constraints such as `constraint_max_dd <= x` and `constraint_sharpe >= y`. This forces the sampler to operate within predefined risk boundaries, acting as a more formal regularizer than the current linear penalty approach. 3. **Rationale:** A composite objective provides a more holistic view of strategy quality, allowing for explicit trade-off management [43]. Using formal constraints is a middle ground between unconstrained optimization and full-blown MOO, offering a cleaner way to enforce risk limits [23]. These enhancements will make the optimization process more targeted and aligned with explicit business requirements.

Phase 3: Long-Term Strategic Shift - Transitioning to Multi-Objective Optimization

After establishing a solid foundation with robust validation and enhanced single-objective functions, the system can evolve towards a more principled approach to handling multiple, conflicting goals. 1. **Action:** Replace the Optuna sampler with an implementation of the **Non-dominated Sorting Genetic Algorithm II (NSGA-II)**, available in libraries like Pymoo [8] [16]. Modify the `objective_wrapper` to return a vector of scores, such as `[psr_score, -max_drawdown, profit_factor]`. 2. **Action:** To manage the high computational cost, implement a **Surrogate-Assisted Evolutionary Algorithm (SAEA)**. Use a Gaussian Process (GP) surrogate model to approximate the expensive backtest evaluations. The NSGA-II would primarily query the cheap surrogate, with the true backtester only being called for a select number of promising candidates identified by an acquisition function [7] [8]. 3. **Rationale:** MOO is the theoretically soundest approach for exploring the inherent trade-offs in strategy design [14]. It produces a Pareto front of optimal compromises, providing a rich set of solutions from which the best one can be chosen based on live market conditions or operational constraints [14]. The use of surrogate models is essential to make this computationally feasible for the GPU-native XAUUSD system [5]. This represents the ultimate evolution of the calibration framework, transforming it from a simple optimizer into a comprehensive strategy exploration engine.

By following this phased roadmap, the team can incrementally build a state-of-the-art calibration pipeline. Starting with the most critical fix—robust validation—and progressively layering in more sophisticated scoring and optimization techniques ensures a managed and effective transition away from the limitations of the CAGR-based objective.

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